

Proof of Proposition 3.11:

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In[*]:= (*We establish the multiplication for SW. We
        set the variable l to be sw. The bar denotes inverse. *)
vars = {s, w, l};
mi,j→k[Z_] := Expand[Z] /. {si sj → 1, wi wj → 0, si wj → lk, wi sj → -lk, li sj → -wk, si lj → wk,
        li wj → 0, wi lj → 0, li lj → 0, x-i /; MemberQ[vars, x] ⇒ xk, x-j /; MemberQ[vars, x] ⇒ xk}}
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In[*]:= (* We start by declaring the elements R and κ: *)
κi := λ1 + λ2 si + λ3 wi + λ4 li;
κ̄i := λ̄1 + λ̄2 si + λ̄3 wi + λ̄4 li;
Ri,j := μ1,1 + μ1,2 si + μ1,3 wi + μ1,4 li + μ2,1 sj + μ2,2 si sj + μ2,3 wi sj + μ2,4 li sj +
        μ3,1 wj + μ3,2 si wj + μ3,3 wi wj + μ3,4 li wj + μ4,1 lj + μ4,2 si lj + μ4,3 wi lj + μ4,4 li lj
R̄i,j := μ̄1,1 + μ̄1,2 si + μ̄1,3 wi + μ̄1,4 li + μ̄2,1 sj + μ̄2,2 si sj + μ̄2,3 wi sj + μ̄2,4 li sj +
        μ̄3,1 wj + μ̄3,2 si wj + μ̄3,3 wi wj + μ̄3,4 li wj + μ̄4,1 lj + μ̄4,2 si lj + μ̄4,3 wi lj + μ̄4,4 li lj
(*We also implement the element v :*)
vi := R1,3 κ̄2 // m1,2→1 // m1,3→i
v̄i := R̄3,1 κ̄2 // m1,2→1 // m1,3→i
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In[*]:= (*By the previous lemma, we simply set the values of the bar variables: *)
{λ̄1, λ̄2, λ̄3, λ̄4} = {λ1 / (λ12 - λ22), -λ2 / (λ12 - λ22), -λ3 / (λ12 - λ22), -λ4 / (λ12 - λ22)};

rulesRinv =
{μ̄1,1 → -((μ1,13 + μ1,1 μ1,22 + μ1,1 μ2,12 - 2 μ1,2 μ2,1 μ2,2 + μ1,1 μ2,22) / (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 - 2 μ1,1 μ2,12 -
        2 μ1,22 μ2,1 + μ2,14 + 8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)),
μ̄1,2 → -((μ1,12 μ1,2 - μ1,23 + μ1,2 μ2,12 - 2 μ1,1 μ2,1 μ2,2 + μ1,2 μ2,22) / (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 -
        2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 + 8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)),
μ̄1,3 → -((μ1,12 μ1,3 - μ1,22 μ1,3 + μ1,3 μ2,12 - μ1,3 μ2,22 - 2 μ1,1 μ2,1 μ2,3 + 2 μ1,2 μ2,2 μ2,3) /
        (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 - 2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 +
        8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)), μ̄1,4 →
        -((μ1,12 μ1,4 - μ1,22 μ1,4 + μ1,4 μ2,12 - μ1,4 μ2,22 - 2 μ1,1 μ2,1 μ2,4 + 2 μ1,2 μ2,2 μ2,4) / (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 -
        2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 + 8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)),
μ̄2,1 → -((μ1,12 μ2,1 + μ1,22 μ2,1 - μ2,13 - 2 μ1,1 μ1,2 μ2,2 + μ2,1 μ2,22) / (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 -
        2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 + 8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)),
μ̄2,2 → -((-2 μ1,1 μ1,2 μ2,1 + μ1,12 μ2,2 + μ1,22 μ2,2 + μ2,12 μ2,2 - μ2,13) / (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 -
        2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 + 8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)),
μ̄2,3 → -((-2 μ1,1 μ1,3 μ2,1 + 2 μ1,2 μ1,3 μ2,2 + μ1,12 μ2,3 - μ1,22 μ2,3 + μ2,12 μ2,3 - μ2,22 μ2,3) /
        (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 - 2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 +
        8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)), μ̄2,4 →
        -((-2 μ1,1 μ1,4 μ2,1 + 2 μ1,2 μ1,4 μ2,2 + μ1,12 μ2,4 - μ1,22 μ2,4 + μ2,12 μ2,4 - μ2,22 μ2,4) / (μ1,14 - 2 μ1,1 μ1,22 + μ1,14 -
        2 μ1,1 μ2,12 - 2 μ1,22 μ2,1 + μ2,14 + 8 μ1,1 μ1,2 μ2,1 μ2,2 - 2 μ1,1 μ2,22 - 2 μ1,22 μ2,22 - 2 μ2,12 μ2,22 + μ2,24)),
μ̄3,1 → -((μ1,12 μ3,1 + μ1,22 μ3,1 - μ2,12 μ3,1 - μ2,22 μ3,1 - 2 μ1,1 μ1,2 μ3,2 + 2 μ2,1 μ2,2 μ3,2) /
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In[*]:= A = {2 λ2 (μ2,4 - μ4,2), -2 λ2 (μ2,3 - μ3,2), -2 λ1 λ2 (-2 λ3 μ4,2 + λ2 (μ3,4 + μ4,3)), -2 λ22 (μ3,2 μ4,1 - μ3,1 μ4,2),
-2 λ2 (μ2,2 μ4,1 - μ2,1 μ4,2), -2 λ1 λ22 (-2 μ3,2 μ4,2 + μ2,2 (μ3,4 + μ4,3)), 2 λ1 λ22 (-2 μ3,1 μ4,2 + μ2,1 (μ3,4 + μ4,3)),
2 λ1 λ22 (-2 μ1,3 μ4,2 + μ1,2 (μ3,4 + μ4,3)), -4 λ12 λ22 (μ3,22 + μ4,22 - μ2,2 (μ3,3 + μ4,4)),
-4 λ12 λ22 (μ3,1 μ3,2 + μ4,1 μ4,2 - μ2,1 (μ3,3 + μ4,4)), -2 λ22 (μ1,4 μ3,2 - μ1,3 μ4,2),
-4 λ12 λ22 (μ1,3 μ3,2 + μ1,4 μ4,2 - μ1,2 (μ3,3 + μ4,4)), -2 λ1 λ2 (-2 λ4 μ3,2 + λ2 (μ3,4 + μ4,3)),
4 λ12 λ2 (λ3 μ3,2 + λ4 μ4,2 - λ2 (μ3,3 + μ4,4)), -2 λ2 (μ2,2 μ3,1 - μ2,1 μ3,2), -2 λ22 (μ1,4 μ3,1 - μ1,3 μ4,1),
-2 λ2 (λ4 μ3,1 - λ3 μ4,1), -2 λ2 (μ1,4 μ2,2 - μ1,2 μ4,2), -2 λ2 (μ1,3 μ2,2 - μ1,2 μ3,2),
2 λ4 μ2,2 - 2 λ2 μ4,2, 2 λ3 μ2,2 - 2 λ2 μ3,2, -2 λ2 (μ1,4 μ2,1 - μ1,2 μ4,1), -2 λ2 (μ1,3 μ2,1 - μ1,2 μ3,1),
2 λ4 μ2,1 - 2 λ2 μ4,1, -2 λ3 μ2,1 + 2 λ2 μ3,1, -2 λ2 (λ4 μ1,3 - λ3 μ1,4), 2 λ4 μ1,2 - 2 λ2 μ1,4,
-2 λ3 μ1,2 + 2 λ2 μ1,3, -2 λ12 λ2 (2 λ4 μ4,22 + λ2 (μ3,2 (μ3,4 + μ4,3) - 2 μ4,2 (μ3,3 + μ4,4))),
2 λ12 λ2 (2 λ4 μ4,1 μ4,2 + λ2 (μ3,1 (μ3,4 + μ4,3) - 2 μ4,1 (μ3,3 + μ4,4))),
2 λ12 λ2 (2 λ4 μ1,4 μ4,2 + λ2 (μ1,3 (μ3,4 + μ4,3) - 2 μ1,4 (μ3,3 + μ4,4))),
-2 λ12 λ2 (2 λ42 μ4,2 + λ2 (λ3 (μ3,4 + μ4,3) - 2 λ4 (μ3,3 + μ4,4))), 4 λ12 λ22
(-(μ3,12 + μ4,12) μ4,2) + μ2,1 μ4,1 (μ3,3 + μ4,4), 4 λ12 λ22 (-(μ1,3 μ3,1 + μ1,4 μ4,1) μ4,2) + μ1,2 μ4,1 (μ3,3 + μ4,4),
-2 λ12 λ22 (μ3,12 (μ3,4 + μ4,3) + μ4,12 (μ3,4 + μ4,3) - 2 μ3,1 μ4,1 (μ3,3 + μ4,4)),
2 λ12 λ24 (μ1,4 μ4,1 (μ3,4 + μ4,3) + μ1,3 (μ3,1 (μ3,4 + μ4,3) - 2 μ4,1 (μ3,3 + μ4,4))),
2 λ12 λ23 (λ4 μ4,1 (μ3,4 + μ4,3) + λ3 (μ3,1 (μ3,4 + μ4,3) - 2 μ4,1 (μ3,3 + μ4,4))),
2 λ12 λ23 (μ1,3 (μ3,4 + μ4,3) + μ1,4 (μ3,4 + μ4,3) - 2 μ1,3 μ1,4 (μ3,3 + μ4,4)),
4 λ12 λ22 (-(μ1,32 + μ1,42) μ4,2) + μ1,2 μ1,4 (μ3,3 + μ4,4),
2 λ12 λ23 (-λ4 μ1,4 (μ3,4 + μ4,3) + λ3 (-μ1,3 (μ3,4 + μ4,3) + 2 μ1,4 (μ3,3 + μ4,4))),
2 λ12 λ22 (λ32 (μ3,4 + μ4,3) + λ42 (μ3,4 + μ4,3) - 2 λ3 λ4 (μ3,3 + μ4,4))};
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(*We start by the first case (R_{1,2}R_{4,3}//m_{2,4→2})-(R_{1,2}R_{4,3}//m_{4,2→2}*)

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In[*]:= vari = {s1, s2, s3, w1, w2, w3, l1, l2, l3};
variL = {"s1", "s2", "s3", "w1", "w2", "w3", "l1", "l2", "l3"};
(*p=(R1,2R4,3//m1,4→1)-(R1,2R4,3//m4,1→1);*)
p = (R1,2 R4,3 // m1,4→1) - (R1,2 R4,3 // m4,1→1);
cr = CoefficientRules[p, vari];
convert[{v___Integer}] := StringJoin@Pick[variL, {v}, 1];
list = (convert[First[#]] → Last[#]) & /@ cr
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Out[*]:= {s2w1w3 → -2 μ2,4 μ3,2 + 2 μ2,2 μ3,4, s2w1l3 → -2 μ2,4 μ4,2 + 2 μ2,2 μ4,4, s2w1 → 2 μ1,4 μ2,2 - 2 μ1,2 μ2,4,
s2w3l1 → -2 μ2,3 μ3,2 + 2 μ2,2 μ3,3, s2l1l3 → -2 μ2,3 μ4,2 + 2 μ2,2 μ4,3, s2l1 → 2 μ1,3 μ2,2 - 2 μ1,2 μ2,3,
s3w1w2 → 2 μ2,4 μ3,2 - 2 μ2,2 μ3,4, s3w1l2 → 2 μ2,4 μ4,2 - 2 μ2,2 μ4,4, s3w1 → -2 μ1,4 μ2,2 + 2 μ1,2 μ2,4,
s3w2l1 → 2 μ2,3 μ3,2 - 2 μ2,2 μ3,3, s3l1l2 → 2 μ2,3 μ4,2 - 2 μ2,2 μ4,3, s3l1 → -2 μ1,3 μ2,2 + 2 μ1,2 μ2,3,
w1w2l3 → -2 μ3,4 μ4,2 + 2 μ3,2 μ4,4, w1w2 → 2 μ1,4 μ3,2 - 2 μ1,2 μ3,4, w1w3l2 → 2 μ3,4 μ4,2 - 2 μ3,2 μ4,4,
w1w3 → -2 μ1,4 μ3,2 + 2 μ1,2 μ3,4, w1l2 → 2 μ1,4 μ4,2 - 2 μ1,2 μ4,4, w1l3 → -2 μ1,4 μ4,2 + 2 μ1,2 μ4,4,
w2l1l3 → -2 μ3,3 μ4,2 + 2 μ3,2 μ4,3, w2l1 → 2 μ1,3 μ3,2 - 2 μ1,2 μ3,3, w3l1l2 → 2 μ3,3 μ4,2 - 2 μ3,2 μ4,3,
w3l1 → -2 μ1,3 μ3,2 + 2 μ1,2 μ3,3, l1l2 → 2 μ1,3 μ4,2 - 2 μ1,2 μ4,3, l1l3 → -2 μ1,3 μ4,2 + 2 μ1,2 μ4,3}
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(*We need to check that the coefficients corresponding with monomials that have no two copies of w,l vanish. We start by listing them*)


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In[*]:= (*λ2=0;λ3=0;λ4=0;*)
eqns1 = {};
vari = {s1, s2, s3, w1, w2, w3, l1, l2, l3};
p = ((R1,2 R4,3 R5,6 // m1,4→1 // m2,5→2 // m3,6→3) -
      (R1,6 R2,3 R4,5 // m1,4→1 // m2,5→2 // m3,6→3));
cr = CoefficientRules[p, vari];
eqns2 = Join[eqns1, Thread[Values[cr] == 0]];
vari = {s1, w1, l1};
p = (R1,3 κ2 // m1,2→1 // m1,3→1) - (R3,1 κ2 // m1,2→1 // m1,3→1);
cr = CoefficientRules[p, vari];
eqns3 = Join[eqns2, Thread[Values[cr] == 0]];
vari = {s1, s2, w1, w2, l1, l2};
p = ((κ1 κ2 R3,4 κ5 κ6 // m1,3→1 // m1,5→1 // m2,4→2 // m2,6→2) - R1,2);
cr = CoefficientRules[p, vari];
eqns4 = Join[eqns3, Thread[Values[cr] == 0]];
vari = {s1, s2, w1, w2, l1, l2};
p = ((R1,4 R3,2 κ6 // m1,3→1 // m2,6→2 // m2,4→2) - κ2);
cr = CoefficientRules[p, vari];
eqns5 = Join[eqns4, Thread[Values[cr] == 0]];

In[*]:= solutionslambda = Solve[(eqns5 /. {λ2 → 0, λ3 → 0, λ4 → 0}),

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$\{\lambda_1, \mu_{1,1}, \mu_{1,2}, \mu_{1,3}, \mu_{1,4}, \mu_{2,1}, \mu_{2,2}, \mu_{2,3}, \mu_{2,4}, \mu_{3,1}, \mu_{3,2}, \mu_{3,3}, \mu_{3,4}, \mu_{4,1}, \mu_{4,2}, \mu_{4,3}, \mu_{4,4}\}$

... Solve: Equations may not give solutions for all "solve" variables.

Out[*]=

$\{\lambda_1 \rightarrow -1, \mu_{1,2} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,2} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,2} \rightarrow 0\},$
 $\{\lambda_1 \rightarrow 1, \mu_{1,2} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,2} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,2} \rightarrow 0\},$
 $\{\lambda_1 \rightarrow -1, \mu_{1,1} \rightarrow 0, \mu_{1,2} \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow \mu_{4,2}, \mu_{3,1} \rightarrow 0, \mu_{4,1} \rightarrow 0\},$
 $\{\lambda_1 \rightarrow 1, \mu_{1,1} \rightarrow 0, \mu_{1,2} \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow \mu_{4,2}, \mu_{3,1} \rightarrow 0, \mu_{4,1} \rightarrow 0\},$
 $\left\{\lambda_1 \rightarrow -1, \mu_{1,1} \rightarrow -\frac{\mu_{2,1}^2}{\mu_{2,2}}, \mu_{1,2} \rightarrow -\mu_{2,1}, \mu_{1,3} \rightarrow -\frac{\mu_{2,1} \mu_{3,2}}{\mu_{2,2}}, \mu_{1,4} \rightarrow -\frac{\mu_{2,1} \mu_{4,2}}{\mu_{2,2}}, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow \mu_{4,2}, \right.$
 $\left. \mu_{3,1} \rightarrow \frac{\mu_{2,1} \mu_{3,2}}{\mu_{2,2}}, \mu_{3,3} \rightarrow \frac{\mu_{3,2}^2 + \mu_{4,2}^2 - \mu_{2,2} \mu_{4,4}}{\mu_{2,2}}, \mu_{4,1} \rightarrow \frac{\mu_{2,1} \mu_{4,2}}{\mu_{2,2}}, \mu_{4,3} \rightarrow \frac{-\mu_{2,2} \mu_{3,4} + 2 \mu_{3,2} \mu_{4,2}}{\mu_{2,2}}\right\},$
 $\left\{\lambda_1 \rightarrow 1, \mu_{1,1} \rightarrow -\frac{\mu_{2,1}^2}{\mu_{2,2}}, \mu_{1,2} \rightarrow -\mu_{2,1}, \mu_{1,3} \rightarrow -\frac{\mu_{2,1} \mu_{3,2}}{\mu_{2,2}}, \mu_{1,4} \rightarrow -\frac{\mu_{2,1} \mu_{4,2}}{\mu_{2,2}}, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow \mu_{4,2}, \right.$
 $\left. \mu_{3,1} \rightarrow \frac{\mu_{2,1} \mu_{3,2}}{\mu_{2,2}}, \mu_{3,3} \rightarrow \frac{\mu_{3,2}^2 + \mu_{4,2}^2 - \mu_{2,2} \mu_{4,4}}{\mu_{2,2}}, \mu_{4,1} \rightarrow \frac{\mu_{2,1} \mu_{4,2}}{\mu_{2,2}}, \mu_{4,3} \rightarrow \frac{-\mu_{2,2} \mu_{3,4} + 2 \mu_{3,2} \mu_{4,2}}{\mu_{2,2}}\right\},$
 $\{\lambda_1 \rightarrow -1, \mu_{1,1} \rightarrow 0, \mu_{1,2} \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow 0, \mu_{3,1} \rightarrow 0, \mu_{4,1} \rightarrow 0, \mu_{4,2} \rightarrow 0\},$
 $\{\lambda_1 \rightarrow 1, \mu_{1,1} \rightarrow 0, \mu_{1,2} \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow 0, \mu_{3,1} \rightarrow 0, \mu_{4,1} \rightarrow 0, \mu_{4,2} \rightarrow 0\},$
 $\left\{\lambda_1 \rightarrow -1, \mu_{1,3} \rightarrow \frac{\mu_{1,2} \mu_{3,2}}{\mu_{2,2}}, \mu_{1,4} \rightarrow \frac{\mu_{1,2} \mu_{4,2}}{\mu_{2,2}}, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow \mu_{4,2}, \mu_{3,1} \rightarrow \frac{\mu_{2,1} \mu_{3,2}}{\mu_{2,2}}, \right.$
 $\left. \mu_{3,3} \rightarrow \frac{\mu_{3,2}^2}{\mu_{2,2}}, \mu_{3,4} \rightarrow \frac{\mu_{3,2} \mu_{4,2}}{\mu_{2,2}}, \mu_{4,1} \rightarrow \frac{\mu_{2,1} \mu_{4,2}}{\mu_{2,2}}, \mu_{4,3} \rightarrow \frac{\mu_{3,2} \mu_{4,2}}{\mu_{2,2}}, \mu_{4,4} \rightarrow \frac{\mu_{4,2}^2}{\mu_{2,2}}\right\},$
 $\left\{\lambda_1 \rightarrow 1, \mu_{1,3} \rightarrow \frac{\mu_{1,2} \mu_{3,2}}{\mu_{2,2}}, \mu_{1,4} \rightarrow \frac{\mu_{1,2} \mu_{4,2}}{\mu_{2,2}}, \mu_{2,3} \rightarrow \mu_{3,2}, \mu_{2,4} \rightarrow \mu_{4,2}, \mu_{3,1} \rightarrow \frac{\mu_{2,1} \mu_{3,2}}{\mu_{2,2}}, \right.$

[illegible]

(*For all cases, the coefficients are zero:*)

$$\text{In}[\bullet]:= \text{equaciones /. solutionslambda /. } \{\lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0\}$$

Out[•]=

[illegible]

(*Now case (b):*)

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In[*]:= eqns1 = {};
vari = {s1, s2, s3, w1, w2, w3, l1, l2, l3};
p = ((R1,2 R4,3 R5,6 // m1,4→1 // m2,5→2 // m3,6→3) -
      (R1,6 R2,3 R4,5 // m1,4→1 // m2,5→2 // m3,6→3));
cr = CoefficientRules[p, vari];
eqns2 = Join[eqns1, Thread[Values[cr] == 0]];
vari = {s1, w1, l1};
p = (R1,3 κ2 // m1,2→1 // m1,3→1) - (R3,1 κ2 // m1,2→1 // m1,3→1);
cr = CoefficientRules[p, vari];
eqns3 = Join[eqns2, Thread[Values[cr] == 0]];
vari = {s1, s2, w1, w2, l1, l2};
p = ((κ1 κ2 R3,4 κ5 κ6 // m1,3→1 // m1,5→1 // m2,4→2 // m2,6→2) - R1,2);
cr = CoefficientRules[p, vari];
eqns4 = Join[eqns3, Thread[Values[cr] == 0]];
vari = {s1, s2, w1, w2, l1, l2};
p = ((R1,4 R3,2 κ6 // m1,3→1 // m2,6→2 // m2,4→2) - κ2);
cr = CoefficientRules[p, vari];
eqns5 = Join[eqns4, Thread[Values[cr] == 0]];

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In[*]:= solutionsmu = Solve[eqns5 /. {μ1,2 → 0, μ2,1 → 0, μ2,2 → 0},
  {λ1, λ2, λ3, λ4, μ1,1, μ1,3, μ1,4, μ2,3, μ2,4, μ3,1, μ3,2, μ3,3, μ3,4, μ4,1, μ4,2, μ4,3, μ4,4}]

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... Solve: Equations may not give solutions for all "solve" variables.

Out[*]=

$$\begin{aligned}
 &\left\{ \lambda_1 \rightarrow -1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow 1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow -1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,1} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow -1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,1} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow 1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,1} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow 1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,1} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \left\{ \lambda_1 \rightarrow -1, \right. \\
 &\quad \left. \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,1} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,1} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow -1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow -\frac{2\mu_{2,3}}{\mu_{1,1}}, \lambda_4 \rightarrow \frac{2\mu_{2,3}}{\mu_{1,1}}, \mu_{1,4} \rightarrow -\mu_{1,3}, \mu_{2,4} \rightarrow -\mu_{2,3}, \mu_{3,1} \rightarrow -\mu_{1,3}, \mu_{3,2} \rightarrow -\mu_{2,3}, \right. \\
 &\quad \left. \mu_{4,1} \rightarrow \mu_{1,3}, \mu_{4,2} \rightarrow \mu_{2,3}, \mu_{4,3} \rightarrow \frac{2\mu_{1,3}^2 - 2\mu_{2,3}^2 - \mu_{1,1}\mu_{3,4}}{\mu_{1,1}}, \mu_{4,4} \rightarrow \frac{-2\mu_{1,3}^2 + 2\mu_{2,3}^2 - \mu_{1,1}\mu_{3,3}}{\mu_{1,1}} \right\}, \\
 &\left\{ \lambda_1 \rightarrow -1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow \frac{2\mu_{2,3}}{\mu_{1,1}}, \lambda_4 \rightarrow \frac{2\mu_{2,3}}{\mu_{1,1}}, \mu_{1,4} \rightarrow \mu_{1,3}, \mu_{2,4} \rightarrow \mu_{2,3}, \mu_{3,1} \rightarrow -\mu_{1,3}, \mu_{3,2} \rightarrow -\mu_{2,3}, \right. \\
 &\quad \left. \mu_{4,1} \rightarrow -\mu_{1,3}, \mu_{4,2} \rightarrow -\mu_{2,3}, \mu_{4,3} \rightarrow \frac{-2\mu_{1,3}^2 + 2\mu_{2,3}^2 - \mu_{1,1}\mu_{3,4}}{\mu_{1,1}}, \mu_{4,4} \rightarrow \frac{-2\mu_{1,3}^2 + 2\mu_{2,3}^2 - \mu_{1,1}\mu_{3,3}}{\mu_{1,1}} \right\}, \\
 &\left\{ \lambda_1 \rightarrow 1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow 0, \lambda_4 \rightarrow 0, \mu_{1,3} \rightarrow 0, \mu_{1,4} \rightarrow 0, \mu_{2,3} \rightarrow 0, \mu_{2,4} \rightarrow 0, \mu_{3,1} \rightarrow 0, \mu_{3,2} \rightarrow 0, \mu_{4,1} \rightarrow 0, \mu_{4,2} \rightarrow 0 \right\}, \\
 &\left\{ \lambda_1 \rightarrow 1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow \frac{2\mu_{2,3}}{\mu_{1,1}}, \lambda_4 \rightarrow -\frac{2\mu_{2,3}}{\mu_{1,1}}, \mu_{1,4} \rightarrow -\mu_{1,3}, \mu_{2,4} \rightarrow -\mu_{2,3}, \mu_{3,1} \rightarrow -\mu_{1,3}, \mu_{3,2} \rightarrow -\mu_{2,3}, \right. \\
 &\quad \left. \mu_{4,1} \rightarrow \mu_{1,3}, \mu_{4,2} \rightarrow \mu_{2,3}, \mu_{4,3} \rightarrow \frac{2\mu_{1,3}^2 - 2\mu_{2,3}^2 - \mu_{1,1}\mu_{3,4}}{\mu_{1,1}}, \mu_{4,4} \rightarrow \frac{-2\mu_{1,3}^2 + 2\mu_{2,3}^2 - \mu_{1,1}\mu_{3,3}}{\mu_{1,1}} \right\}, \\
 &\left\{ \lambda_1 \rightarrow 1, \lambda_2 \rightarrow 0, \lambda_3 \rightarrow -\frac{2\mu_{2,3}}{\mu_{1,1}}, \lambda_4 \rightarrow -\frac{2\mu_{2,3}}{\mu_{1,1}}, \mu_{1,4} \rightarrow \mu_{1,3}, \mu_{2,4} \rightarrow \mu_{2,3}, \mu_{3,1} \rightarrow -\mu_{1,3}, \mu_{3,2} \rightarrow -\mu_{2,3}, \right.
 \end{aligned}$$

[illegible]

(*We again have a finite number of cases. For all of them, the coefficients are zero:*)

$$ln[*]:= \text{ecuaciones /. solutionsmu /. } \{\mu_{1,2} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,2} \rightarrow 0\}$$

Out[•]=

[illegible]

ALL CASES:

$$\begin{aligned}
& \left\{ \left\{ 0, \frac{\mu_{1,2} (\mu_{1,1}^2 - \mu_{1,2}^2 + \mu_{2,1}^2 - \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{\mu_{1,2} \mu_{2,1}}{\lambda_2}, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{-\mu_{1,1}^2 + \mu_{1,2}^2 + \mu_{2,1}^2}{\lambda_2}, 0, 0, 0, -\frac{\mu_{2,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ \frac{2 \mu_{1,2} (-\mu_{1,1} \mu_{2,1} + \mu_{1,2} \mu_{2,2})}{\lambda_2}, 0, 0, 0, -\frac{2 \mu_{1,1} \mu_{1,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{2 \mu_{1,2} \mu_{2,2}}{\lambda_2}, 0, 0, 0, -\frac{2 \mu_{1,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ 0, \frac{2 \mu_{1,2} (\mu_{1,1} \mu_{2,1} - \mu_{1,2} \mu_{2,2})}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{2 \mu_{1,1} \mu_{1,2}}{\lambda_2}, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{2 \mu_{1,2} \mu_{2,2}}{\lambda_2}, 0, 0, 0, -\frac{2 \mu_{1,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ \left\{ \frac{2 \mu_{1,1} \mu_{1,2} \mu_{2,1} - \mu_{1,1}^2 \mu_{2,2} + \mu_{2,2} (-\mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, \frac{2 \mu_{1,1} \mu_{1,2} - 2 \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, \right. \right. \right. \\
& \quad \left. \left. 0, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ 0, \frac{-2 \mu_{1,1} \mu_{1,2} \mu_{2,1} + \mu_{1,1}^2 \mu_{2,2} + \mu_{2,2} (\mu_{1,2}^2 + \mu_{2,1}^2 - \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{2 \mu_{1,1} \mu_{1,2} - 2 \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0 \right\}, \right. \\
& \quad \left. 0 \right\}, \left\{ \left\{ 0, 0, 0, 0, -\frac{\mu_{1,2} \mu_{2,1}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,2}^2}{\lambda_2}, 0, 0, \right. \right. \\
& \quad \left. \left. 0, \frac{-2 \mu_{1,1} \mu_{1,2} + \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{\mu_{1,2} \mu_{2,1}}{\lambda_2}, 0, 0, 0, \frac{\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,2}^2}{\lambda_2}, 0, 0, \right. \right. \\
& \quad \left. \left. 0, \frac{-2 \mu_{1,1} \mu_{1,2} + \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ \frac{-\mu_{1,1}^2 \mu_{2,1} + 2 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (-\mu_{1,2}^2 + \mu_{2,1}^2 - \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, \frac{-\mu_{1,1}^2 - \mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2}{\lambda_2}, 0, 0, 0, 0, \right. \right. \\
& \quad \left. \left. 0, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ 0, \frac{\mu_{1,1}^2 \mu_{2,1} - 2 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (\mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{-\mu_{1,1}^2 - \mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2}{\lambda_2}, 0 \right\}, \right. \\
& \quad \left. 0 \right\}, \left\{ \left\{ 0, 0, 0, 0, \frac{2 \mu_{1,2} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{2 \mu_{1,1} \mu_{1,2}}{\lambda_2}, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2}{\lambda_2}, 0 \right\}, 0 \right\}, \\
& \left\{ \left\{ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{2 \mu_{1,2} \mu_{2,2}}{\lambda_2}, 0, 0, 0, -\frac{2 \mu_{1,1} \mu_{1,2}}{\lambda_2}, 0, 0, 0, \right. \right. \\
& \quad \left. \left. \frac{\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\},
\end{aligned}$$

$$\begin{aligned} & \left\{ \left\{ \left\{ 0, 0, 0, 0, \frac{2 \mu_{1,1} \mu_{1,2} - 2 \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{2 \mu_{1,1} \mu_{1,2} - 2 \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, \frac{\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,1}^2 - \mu_{2,2}^2}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,1}^2 - \mu_{2,2}^2}{\lambda_2}, 0, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, \frac{-\mu_{1,1}^2 - \mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{-\mu_{1,1}^2 - \mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2}{\lambda_2}, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, \frac{-2 \mu_{1,1} \mu_{1,2} + 2 \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{-2 \mu_{1,1} \mu_{1,2} + 2 \mu_{2,1} \mu_{2,2}}{\lambda_2}, 0, 0, 0, 0, \right. \right. \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}\}, \\ & \left\{ \left\{ -\frac{\mu_{1,2} (\mu_{1,1}^4 + \mu_{1,2}^4 + 8 \mu_{1,1} \mu_{1,2} \mu_{2,1} \mu_{2,2} + (\mu_{2,1}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,2}^2 (\mu_{2,1}^2 + \mu_{2,2}^2) - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2))}{\lambda_2}, \right. \right. \\ & \quad 0, 0, 0, \frac{\mu_{1,2} (2 \mu_{1,1}^2 \mu_{2,1} - 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (2 \mu_{1,2}^2 - \mu_{2,1}^2 + 2 \mu_{2,2}^2))}{\lambda_2}, 0, \\ & \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{\mu_{1,1}^4 + (\mu_{1,2}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,2}^2)}{\lambda_2}, \\ & \quad 0, 0, 0, \frac{\mu_{2,2} (2 \mu_{1,1}^2 \mu_{2,1} - 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (2 \mu_{1,2}^2 - \mu_{2,1}^2 + 2 \mu_{2,2}^2))}{\lambda_2}, \\ & \quad \left. \left. 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right\}, 0 \right\}, \\ & \left\{ \left\{ 0, \frac{\mu_{1,2} (\mu_{1,1}^4 + \mu_{1,2}^4 + 8 \mu_{1,1} \mu_{1,2} \mu_{2,1} \mu_{2,2} + (\mu_{2,1}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,2}^2 (\mu_{2,1}^2 + \mu_{2,2}^2) - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2))}{\lambda_2}, \right. \right. \\ & \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{\mu_{1,2} (2 \mu_{1,1}^2 \mu_{2,1} - 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (2 \mu_{1,2}^2 - \mu_{2,1}^2 + 2 \mu_{2,2}^2))}{\lambda_2}, \\ & \quad \left. \left. 0, 0, 0, 0, -\frac{\mu_{1,1}^4 + (\mu_{1,2}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, \right. \right. \end{aligned}$$

$$\begin{aligned}
& \left\{ \left\{ \frac{\mu_{1,2} (\mu_{1,1}^4 + \mu_{1,2}^4 + 8 \mu_{1,1} \mu_{1,2} \mu_{2,1} \mu_{2,2} + (\mu_{2,1}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,2}^2 (\mu_{2,1}^2 + \mu_{2,2}^2) - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2))}{\lambda_2}, \right. \right. \\
& \quad 0, 0, 0, \frac{\mu_{1,2} (-2 \mu_{1,1}^2 \mu_{2,1} + 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (-2 \mu_{1,2}^2 + \mu_{2,1}^2 - 2 \mu_{2,2}^2))}{\lambda_2}, 0, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{\mu_{1,1}^4 + (\mu_{1,2}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,2}^2)}{\lambda_2}, \\
& \quad 0, 0, 0, \frac{\mu_{2,2} (-2 \mu_{1,1}^2 \mu_{2,1} + 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (-2 \mu_{1,2}^2 + \mu_{2,1}^2 - 2 \mu_{2,2}^2))}{\lambda_2}, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \}, 0 \}, \\
& \left\{ \left\{ 0, -\frac{\mu_{1,2} (\mu_{1,1}^4 + \mu_{1,2}^4 + 8 \mu_{1,1} \mu_{1,2} \mu_{2,1} \mu_{2,2} + (\mu_{2,1}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,2}^2 (\mu_{2,1}^2 + \mu_{2,2}^2) - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,1}^2 + \mu_{2,2}^2))}{\lambda_2}, \right. \right. \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \frac{\mu_{1,2} (-2 \mu_{1,1}^2 \mu_{2,1} + 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (-2 \mu_{1,2}^2 + \mu_{2,1}^2 - 2 \mu_{2,2}^2))}{\lambda_2}, \\
& \quad 0, 0, 0, \frac{\mu_{1,1}^4 + (\mu_{1,2}^2 - \mu_{2,2}^2)^2 - 2 \mu_{1,1}^2 (\mu_{1,2}^2 + \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, 0, \\
& \quad \frac{\mu_{2,2} (-2 \mu_{1,1}^2 \mu_{2,1} + 8 \mu_{1,1} \mu_{1,2} \mu_{2,2} + \mu_{2,1} (-2 \mu_{1,2}^2 + \mu_{2,1}^2 - 2 \mu_{2,2}^2))}{\lambda_2}, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \}, 0 \}, \\
& \left\{ \left\{ \frac{1}{\lambda_2} (-2 \mu_{1,1}^3 \mu_{1,2} \mu_{2,1} + \mu_{1,1}^4 \mu_{2,2} + 2 \mu_{1,1}^2 \mu_{2,2} (\mu_{2,1}^2 - \mu_{2,2}^2) + 2 \mu_{1,1} \mu_{1,2} \mu_{2,1} (\mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2) + \right. \right. \\
& \quad \mu_{2,2} (-\mu_{1,2}^4 + (\mu_{2,1}^2 - \mu_{2,2}^2)^2)), 0, 0, 0, \frac{2 (2 \mu_{1,1} (\mu_{1,2}^3 - \mu_{1,2} \mu_{2,1}^2) + \mu_{1,1}^2 \mu_{2,1} \mu_{2,2} + \mu_{2,1} \mu_{2,2} (\mu_{2,1}^2 - \mu_{2,2}^2))}{\lambda_2}, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{2 \mu_{1,2} \mu_{2,2} (\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,2}^2)}{\lambda_2}, 0, 0, 0, 0, \\
& \quad \frac{2 (-2 \mu_{1,1}^2 \mu_{1,2} \mu_{2,1} + \mu_{1,1}^3 \mu_{2,2} - \mu_{1,2} \mu_{2,1} \mu_{2,2}^2 + \mu_{1,1} \mu_{2,2} (3 \mu_{1,2}^2 + \mu_{2,1}^2 - \mu_{2,2}^2))}{\lambda_2}, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \}, 0 \}, \\
& \left\{ \left\{ 0, \frac{1}{\lambda_2} (2 \mu_{1,1}^3 \mu_{1,2} \mu_{2,1} - \mu_{1,1}^4 \mu_{2,2} - 2 \mu_{1,1} \mu_{1,2} \mu_{2,1} (\mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2) + \mu_{1,1}^2 (-2 \mu_{2,1}^2 \mu_{2,2} + 2 \mu_{2,2}^3) - \right. \right. \\
& \quad \mu_{2,2} (-\mu_{1,2}^4 + (\mu_{2,1}^2 - \mu_{2,2}^2)^2)), 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, \\
& \quad \frac{2 (2 \mu_{1,1} (\mu_{1,2}^3 - \mu_{1,2} \mu_{2,1}^2) + \mu_{1,1}^2 \mu_{2,1} \mu_{2,2} + \mu_{2,1} \mu_{2,2} (\mu_{2,1}^2 - \mu_{2,2}^2))}{\lambda_2}, 0, 0, 0, -\frac{2 \mu_{1,2} \mu_{2,2} (\mu_{1,1}^2 + \mu_{1,2}^2 - \mu_{2,2}^2)}{\lambda_2}, \\
& \quad 0, 0, 0, \frac{2 (-2 \mu_{1,1}^2 \mu_{1,2} \mu_{2,1} + \mu_{1,1}^3 \mu_{2,2} - \mu_{1,2} \mu_{2,1} \mu_{2,2}^2 + \mu_{1,1} \mu_{2,2} (3 \mu_{1,2}^2 + \mu_{2,1}^2 - \mu_{2,2}^2))}{\lambda_2}, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \}, 0 \}, \\
& \left\{ \left\{ \frac{2 (-\mu_{1,1}^3 \mu_{2,1} \mu_{2,2} + \mu_{1,2} \mu_{2,2}^2 (\mu_{1,2}^2 + \mu_{2,1}^2 - \mu_{2,2}^2) + \mu_{1,1} \mu_{2,1} \mu_{2,2} (-3 \mu_{1,2}^2 - \mu_{2,1}^2 + \mu_{2,2}^2) + \mu_{1,1}^2 \mu_{1,2} (2 \mu_{2,1}^2 + \mu_{2,2}^2))}{\lambda_2}, \right. \right. \\
& \quad 0, 0, 0, \frac{4 \mu_{1,1}^2 \mu_{1,2} \mu_{2,1} - 4 \mu_{1,1} (\mu_{1,2}^2 + \mu_{2,1}^2) \mu_{2,2} + \mu_{1,2} \mu_{2,1} (-2 \mu_{1,2}^2 + \mu_{2,1}^2 + 4 \mu_{2,2}^2)}{\lambda_2}, \\
& \quad 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, -\frac{\mu_{1,1}^4 - \mu_{1,2}^4 - 2 \mu_{1,1}^2 \mu_{2,2}^2 + \mu_{2,2}^4}{\lambda_2}, 0, 0, 0, 0,
\end{aligned}$$

[illegible]

```
In[*]:= totalecuaciones /. solutionslambda /. {λ2 → 0, λ3 → 0, λ4 → 0} // Simplify
Out[*]=
```

[illegible]

[illegible]
$$ln[*]:= \text{totalecuaciones} /. \text{solutionsmu} /. \{\mu_{1,2} \rightarrow 0, \mu_{2,1} \rightarrow 0, \mu_{2,2} \rightarrow 0\} // \text{Simplify}$$

Out[•]=

[illegible]

